AICTE PRAGATI-NAV: Project Overview & Core Architecture

Executive Technical & Regulatory Summary - SIH Platform Submission

Project Mission Statement:

To develop an integrated, secure and configurable digital platform for end-to-end automation and monitoring of AICTE approval workflows: application routing, scrutiny stages, EVC scheduling, pendency monitoring, time-bound undertakings, institutional compliance, and document validity.

1. Core Problem Statement & Key Challenges Solved

- **Manual Application Routing:**
  Applications previously delayed between Scrutiny, EVC, and SAC stages due to manual physical file movement.
- **Delayed Expert Visit (EVC) Scheduling:**
  Lack of automated conflict-of-interest checks and delayed 48-hour post-inspection report submissions.
- **Unmonitored Time-Bound Undertakings:**
  Conditional approvals granted without automated countdown tracking or evidence verification workflows.
- **Lapsed 30-Year Land Leases & Fire NOCs:**
  Campuses operating with expired lease deeds, creating acute statutory and legal liabilities.
- **Absence of Automated Multi-Tier Escalations:**
  Delays accumulating across regions without automated intervention by Regional Officers and Bureau Heads.

2. Module A: End-to-End Approval Workflow Automation

The system automatically routes technical institution applications through 5 statutory APH routes:

Route Category	Applicability & Prescribed Stages	Statutory SLA
Only Scrutiny	Standard Extension of Approval (EoA) without intake changes	7 Days
Scrutiny + EVC	Mandatory for Increase in Intake, New Courses, & New Technical Institutions	7d + 2d EVC
Scrutiny + EVC + SAC	Mandatory for Change of Site, Course Closures, or disputed appeals	14 Days SAC
Only EVC (Surprise/Compliance)	Unannounced compliance verification and infrastructure re-inspections	5 Days EVC
EVC + SAC/SHC	Post-rejection appeals, punitive review hearings, and intake restoration	14 Days SAC

3. Milestone Stepper & Authorized Deficiency Revert Workflow

- Automatic Movement: Upon completion of preliminary scrutiny or EVC recommendations, applications auto-transition to downstream stages.
- Mandatory Deficiency Revert: Regional Officers can return applications for deficiency rectification with prescribed APH clauses (e.g. Clause 6.4 Library, Clause 7.12 Faculty Ratio, Clause 2.4 Fire Safety). Automatic 7-day cure window countdown clock commences upon issuance.
- Electronic Case Summary: Instant printable official dossier generated with AICTE emblem, QR verification badge, and digital audit hashes.

Pendency Monitoring, EVC Scheduling & Institutional Compliance

Integrated Zonal Radar, 48h Inspection Dossier & Expiry Matrix

4. Module B: Integrated Dashboard & Regional Pendency Monitoring

PRAGATI-NAV aggregates real-time application movement across all 8 AICTE Regional Offices (WRO, NRO, SRO, CRO, ERO, SCRO, NWRO, SWRO).

Pan-India Zonal Workload & Bottleneck Metrics:

- Western Region (WRO, Mumbai): 2 active cases | Avg Aging: 5.2 days | Status: Optimal Workload
- South-Central (SCRO, Hyderabad): 1 critical case | SLA Breached by 11 days | Status: High Bottleneck (Level 3 Escalated)
- Eastern Region (ERO, Kolkata): 1 active case | Land Lease expiring in 42 days | Status: Urgent Action Watchlist

5. Module C: Expert Visit Committee (EVC) Scheduling & 48-Hour Report Tracking

- Automated EVC Queuing: Applications clearing Preliminary Scrutiny are automatically placed in the EVC Scheduling pool.
- 3-Expert Committee Composition: Mandatory composition of 1 Senior Academician (IIT/NIT/Univ), 1 Architect/Civil Safety Expert, and 1 AICTE RO Repres
- Conflict-of-Interest Validation: Autonomous cross-check verifying no committee member has prior employment or familial affiliation with applicant.
- 48-Hour Report Upload Clock: Strict countdown timer enforced upon visit completion for geo-tagged inspection dossier and digital sign-off.

6. Module D: Time-Bound Undertakings & Compliance Ledger

Certain technical approvals are granted subject to time-bound undertakings (e.g. cadre-ratio recruitment, laboratory equipment, fire NOCs):

Undertaking ID	Institution	Prescribed Condition	Deadline	Status
UND-WRO-01	MIT & Research Pune	Appointment of 2 Professors in AI cadre (7th CPC)	2026-12-15	Pending Evidence
UND-NRO-02	Quantum Inst. Noida	Submission of Occupancy Certificate from GNIDA	2026-11-20	Submitted (Review)
UND-SWRO-01	Bangalore Tech Academy	Installation of Solar Rooftop PV Plant (100 kVA)	2026-11-30	Verified & Closed
UND-SCRO-01	Telangana Inst. Applied Sci	Deposit of Rs. 50 Lakhs in Joint Endowment Fund	2026-10-15	Overdue (SAC Escalated)

7. Module E: 30-Year Land Lease & Document Validity Radar

Continuous validity monitoring for 30-year registered land lease deeds, building safety certificates, and fire safety NOCs:

- Tier 1 (>180 Days): Compliant & Active status. No action required.
- Tier 2 (90-180 Days): 6-Month advance renewal notice automatically dispatched to Principal and Governing Board.
- Tier 3 (<= 90 Days): Critical Renewal Notice. Warns institution that failure to upload registered extension deed freezes subsequent EoA.
- Tier 4 (Expired / Non-Renewed): Automatic regulatory block preventing issuance of Approval Orders until cleared by Regional Officer.

Role-Based Access Control (RBAC) & Governance Layer

Typed Permission Matrix, Data Scoping & Dedicated Role Portals

8. The Three Core AICTE Administrative Personas

The platform implements strict role-based access control governed by /lib/permissions.ts as the single source of truth:

1. Institution (Applicant)

Persona: Dr. Meera Sen, Principal
MIT & Research (ID: 1-748920141)

- Sees ONLY own applications & proofs
- Respond to deficiency notices
- Upload compliance evidence
- Upload renewed lease / Fire NOC
- Track milestone progress bar

Sidebar: Dashboard, My Applications, My Undertakings, My Documents, Alerts

2. Regional Officer (RO)

Persona: Dr. R.K. Patil, RO WRO
AICTE Western Regional Office

- Sees ONLY applications in region
- Advance stage or revert with memo
- Schedule 3-member EVC committee
- Verify / reject undertakings
- Validate renewed lease deeds

Sidebar: Dashboard, Applications, EVC, Undertakings, Docs, Alerts, Reports, Audit

3. Bureau Head (HQ)

Persona: Shri A.K. Sharma, Adviser
Approval Bureau, AICTE New Delhi

- Unrestricted Pan-India (8 Regions)
- 8-region bottleneck heatmap
- Handle Tier-3 apex escalations
- Reassign cases across regions
- Configure APH handbook SLAs

Sidebar: National Dashboard, All Apps, Escalations, Reports, Audit, Rules, Integrations

9. Simplified 3-Tier Escalation Ladder

When application movement or institutional compliance breaches SLA thresholds, the autonomous watchdog fires alerts in order:

Level 1: Institution Reminder
- Trigger: SLA + 2 Days or <= 7d to deadline
- Action: Automated SMS/Email to Principal

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Level 2: Regional Officer Alert
- Trigger: SLA + 5 Days overdue
- Action: Directive memo to RO to reassign or resolve

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Level 3: Bureau Head Apex
- Trigger: SLA + 8 Days overdue
- Action: Emergency docket to Member Secretary

10. Permission UX Rules & Disabled Actions Policy

- Navigation Isolation: Unauthorized sidebar and menu items are completely hidden from the user interface.
- Disabled Buttons with Tooltips: Visible-but-restricted buttons show disabled styling with descriptive tooltips (e.g. "Only Regional Officers can verify docume
- Demo 3-Card Role Selector: Switch roles anytime via the header badge or sidebar to evaluate all three administrative perspectives.

Technical Stack, Government Integrations & Impact

E-Governance Interoperability, Cryptographic Audit & Statutory Disclaimer

11. Central E-Governance API Bridges (Bureau Head View)

PRAGATI-NAV connects directly with 6 central e-governance systems to verify institutional credentials without manual paper verification:

AICTE Central MIS & Approval Portal Bidirectional sync with primary technical institution database, course matrix, and seat intake.	**Active (34ms)**
Bhulekh / NLRMP (National Land Records) Automated digital verification of 30-year registered land lease deeds and non-encumbrance status.	**Connected (58ms)**
DigiLocker NAD (National Academic Depository) Real-time authentication of faculty postgraduate & PhD degrees under AICTE 7th CPC norms.	**Connected (41ms)**
State Fire & Emergency Services Gateway Webhook ingestion for building fire safety NOC approvals, inspection renewals, and validity.	**Connected (72ms)**
NIC / CDAC SMS & WhatsApp Statutory Gateway High-throughput delivery of 30d/15d/7d compliance countdowns and Level 1/2/3 escalation notices.	**Operational (26ms)**
National Single Window System (NSWS) Inter-ministerial data bridge with Ministry of Education, State DTEs, and affiliated Universities.	**Connected (65ms)**

12. Measurable Outcomes & Hackathon Impact Metrics

80% — **Reduction in Follow-up**
Predictive reminders and standardized case summaries elimi
e calls and email follow-ups between Regional Offices and ins

100% — **Lease Expiry Elimination**
180d and 90d advance validity radar prevents unmonitored land lease
ions, guaranteeing legal campus title.

48 Hrs — **EVC Inspection Report SLA**
Strict countdown clock ensures EVC teams upload geo-tagge
ion reports within 48 hours of visit completion.

SHA-256 — **Tamper-Evident Audit Ledger**
Every stage transition, deficiency issuance, and evidence verification
ryptographically signed and permanently logged.

AICTE STATUTORY REGULATORY SAFEGUARD:

The system functions in strict accordance with the applicable AICTE regulations, Approval Process Handbook (APH), and authorized administrative procedures. Automated alerts, recommendations, and workflow transitions assist the approval process but shall not replace statutory scrutiny, physical inspections, or regulatory decisions reserved for competent Council authorities.

Cryptographic System Verification Hash:
SHA256: 9e41b7d8019a284eef1092847a9831cb77019284fa88c12a
Platform Engine: AICTE PRAGATI-NAV v2.6-SIH National Release

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